NETWORKING

DPU Sled Design Standard in a Modular SmartSwitch By Cisco & AMD

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OCP GOLD Member

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Agenda

Cisco will open source the design of a DPU-sled for a modular Smart Switch platform

This will include the following:

- Mechanical form factor, including tray, front panel attachments to the platform, etc.
- PCB outlines and mechanical constraints
- Power delivery specification
- Management signals, PCIe, SPI, I2C, Low Speed signals, and JTAG
- Platform Software APIs

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Motivation

- DPU prevalence is increasing, and a SmartSwitch can subsume DPUs
- Enable deploy ability in-network, bump-in-wire options for DC/T0/T1 layers
- Unified platform – NPU and DPU
  - Divide and Conquer scenarios, NPU:DPU oversubscription
- Unified Management Benefits (platform, DPU, HA, NPU as steering plane)
- Broaden Use cases, Eco-system – switch & DPU sled
- Standardize on software APIs – Platform & Services
- Sizeable multi-billion dollar TAM, ~167% CAGR 2026-2030, >7B by 2030

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Smart Switch – Typical platform





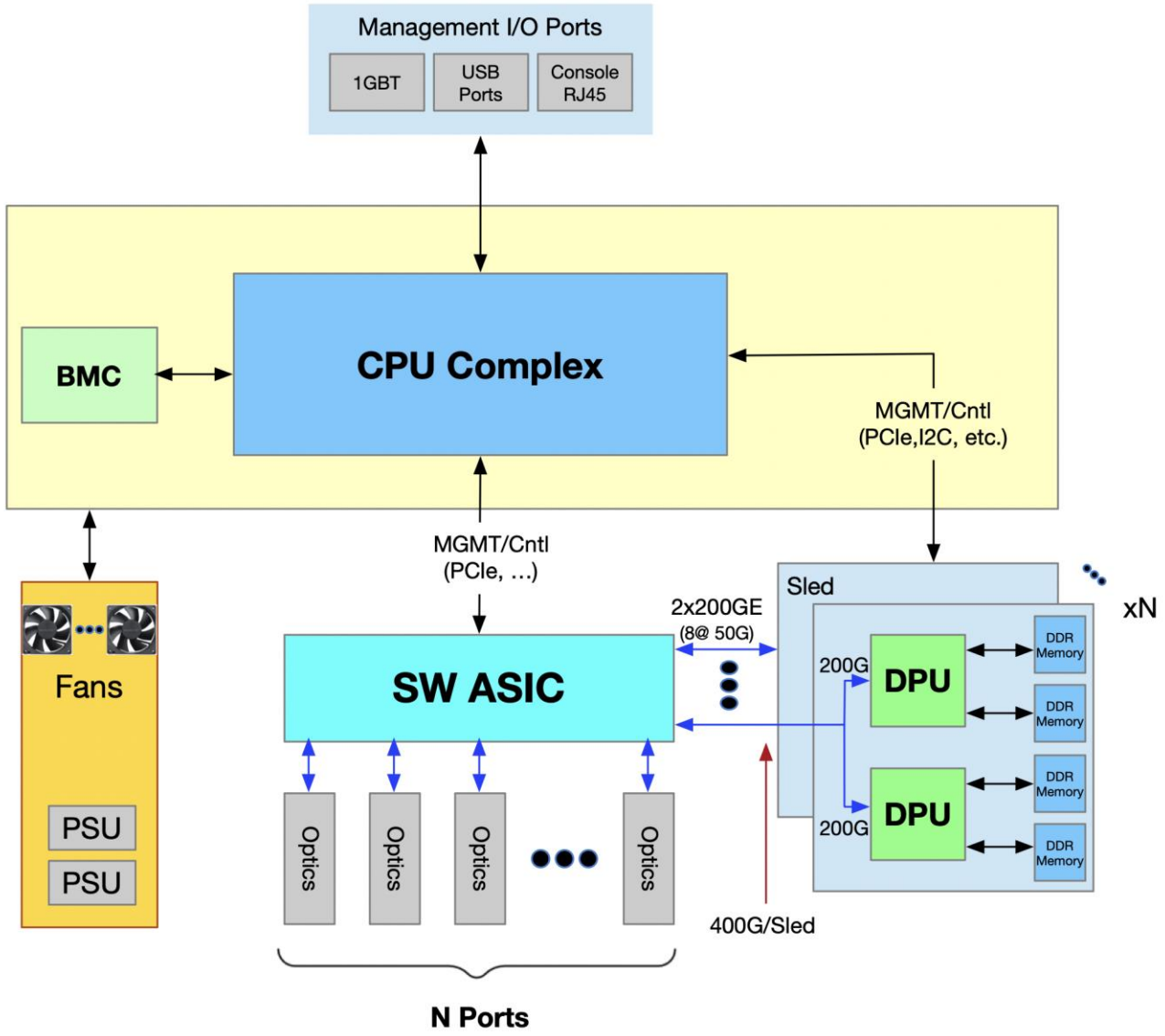
System Overview

Eg: Cisco 8102-28FH-DPU-O - This is a first-generation smart switch equipped with:

- Cisco Silicon One Q200L switching processor
- AMD Elba Data Processing Unit (DPU)
- Housed in a 2 RU 19” EIA rack mountable
- System offers total network bandwidth of 12.8Tbps
- 11.2Tbps for switching and 1.6Tbps for DPU processing
- 28 QSFP-DD 400 GbE ports
- x86 CPU D2779 16C, 128GB DDR4
- DPU has 32GB of DDR4 memory, and 64GB eMMC
- 4 Fan Trays in the back and 2 PSUs (2.3KW ea)

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Representative Smart Switch Block Diagram View

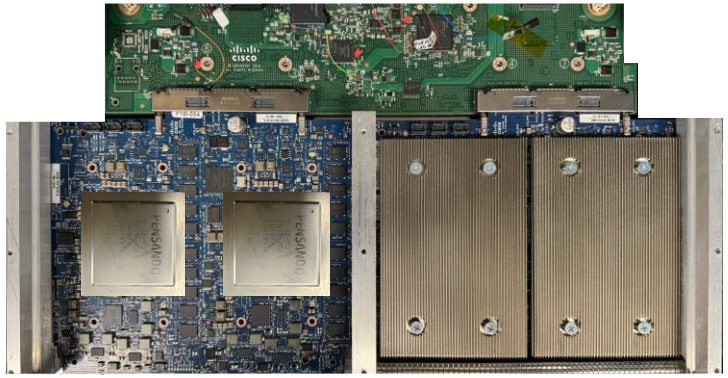


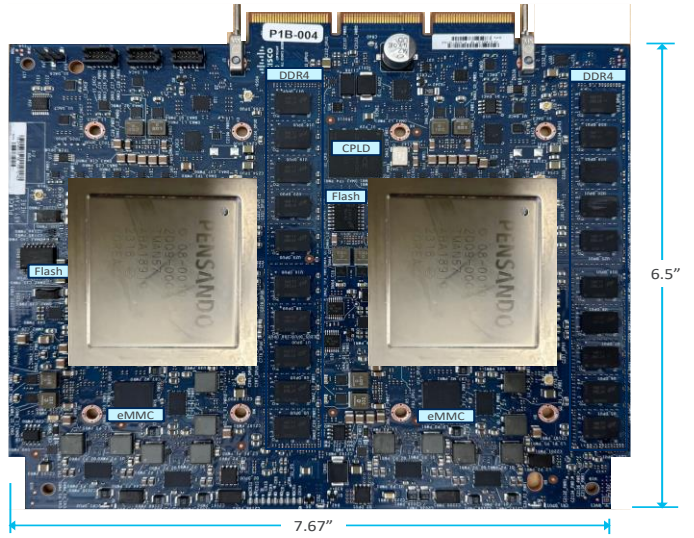
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DPU Sled Overview

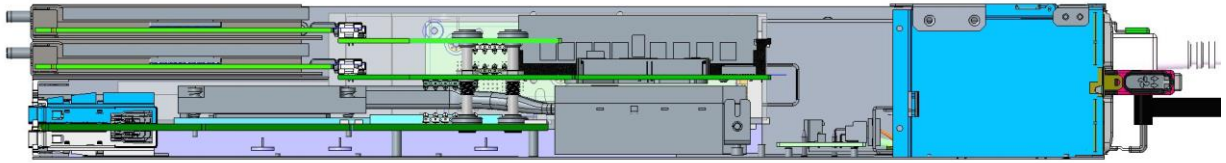
- Up to two DPUs per Sled
- PCIe 4.0 Ports x2
- Ethernet Ports from switch to DPUs
  - 400G BW per DPU slot (8x 50G)
- Max Power per sled - 180W
- PCB Edge Connector:
  - Gold fingers on Sled and Molex SpeedEdge™ connector on host

Internal Top View



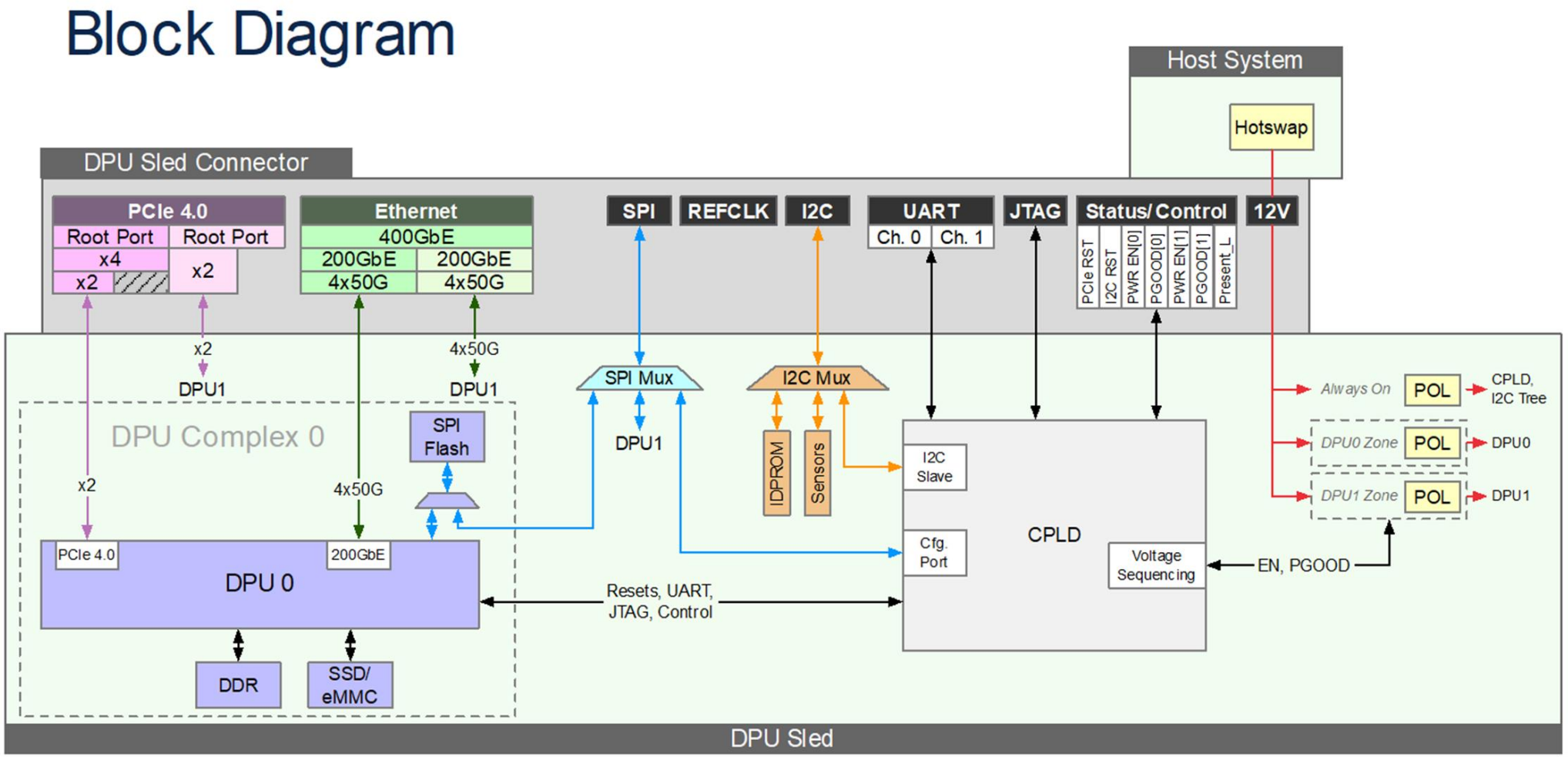


DPU sleds



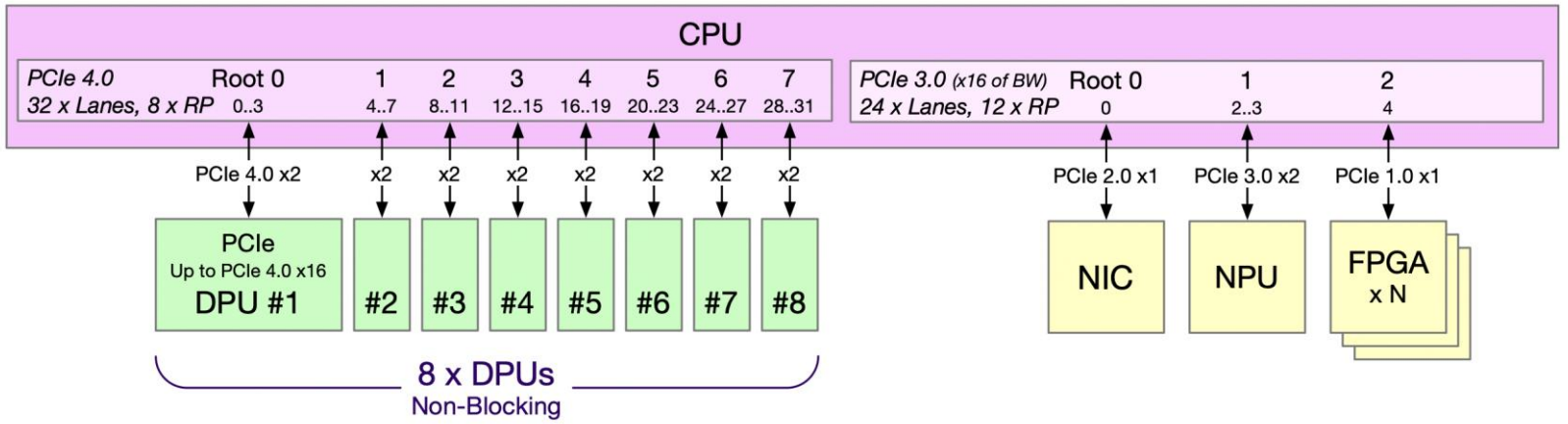
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SLED Interface Block Diagram



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SLED Interface Block Diagram



- PCIe Gen-4 @ 16.0GT/s
- X2 PCIe interface connected to each DPU
- 100MHz PCIe reference clock sourced by the host system
- Support PCIE_PERST
- Support PCIe MSI

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DPU Control, Power, Signals, Thermal

DPU uses edge PCB connector with the following groups of signals and power:

- Power
  - 12V @180W Max switched power by the main board, protected by Hotswap circuit
    - DPU to handle its own power on and shutdown sequence
- Low speed side-band mgmt. interfaces [i2c, uart etc]
- High speed signals, Ethernet 50G per lane

A complete signal pin list at the Speed Edge connector will be provide in a follow-on spec with signal definition and specifics

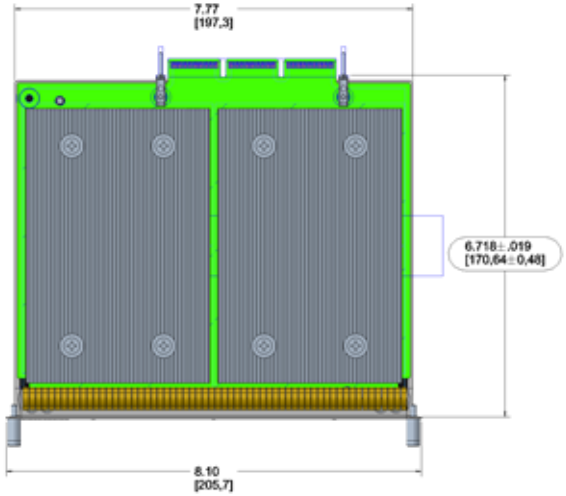
Environmental :

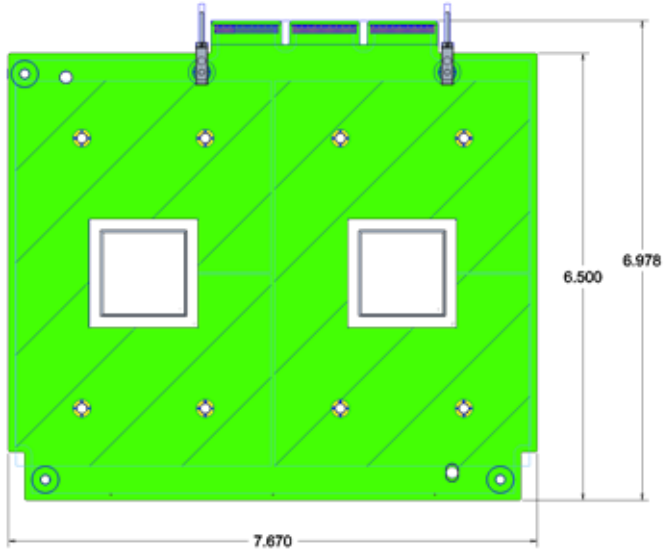
- Operating temperature 0 to 40C
- Non-Op Temp Range: -40 to 70C
- Relative Humidity: 5 to 95% non-condensing
- Maximum air flow thru SLED is 30 CFM

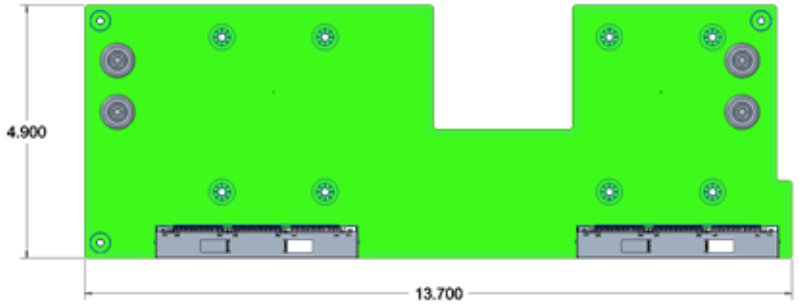
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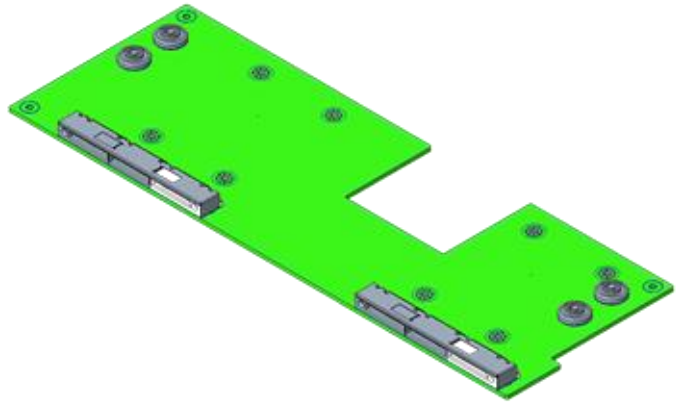
DPU Sled Mechanical Overview

- PCB – 7.670”x6.978”x0.093”
- 2 DPU ASICs



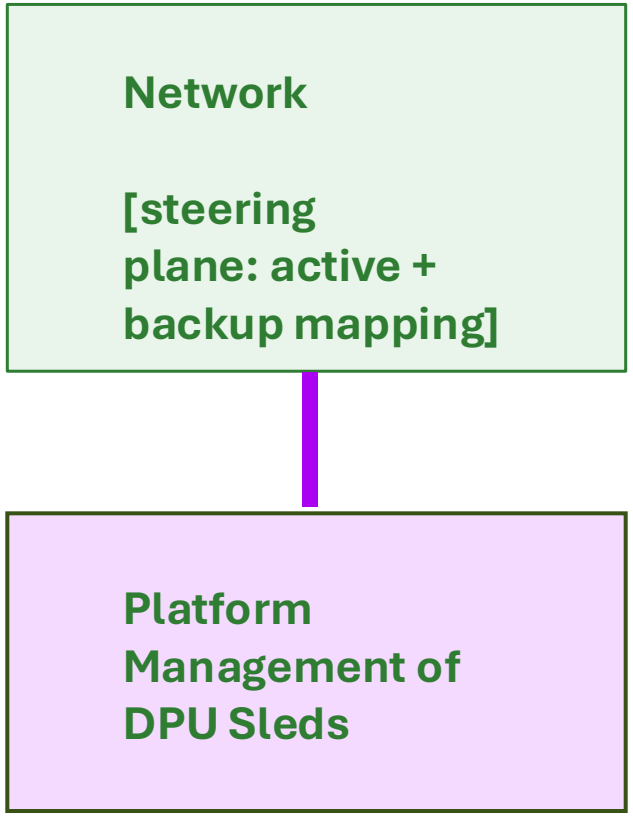






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Software Aspects for DPU Sleds

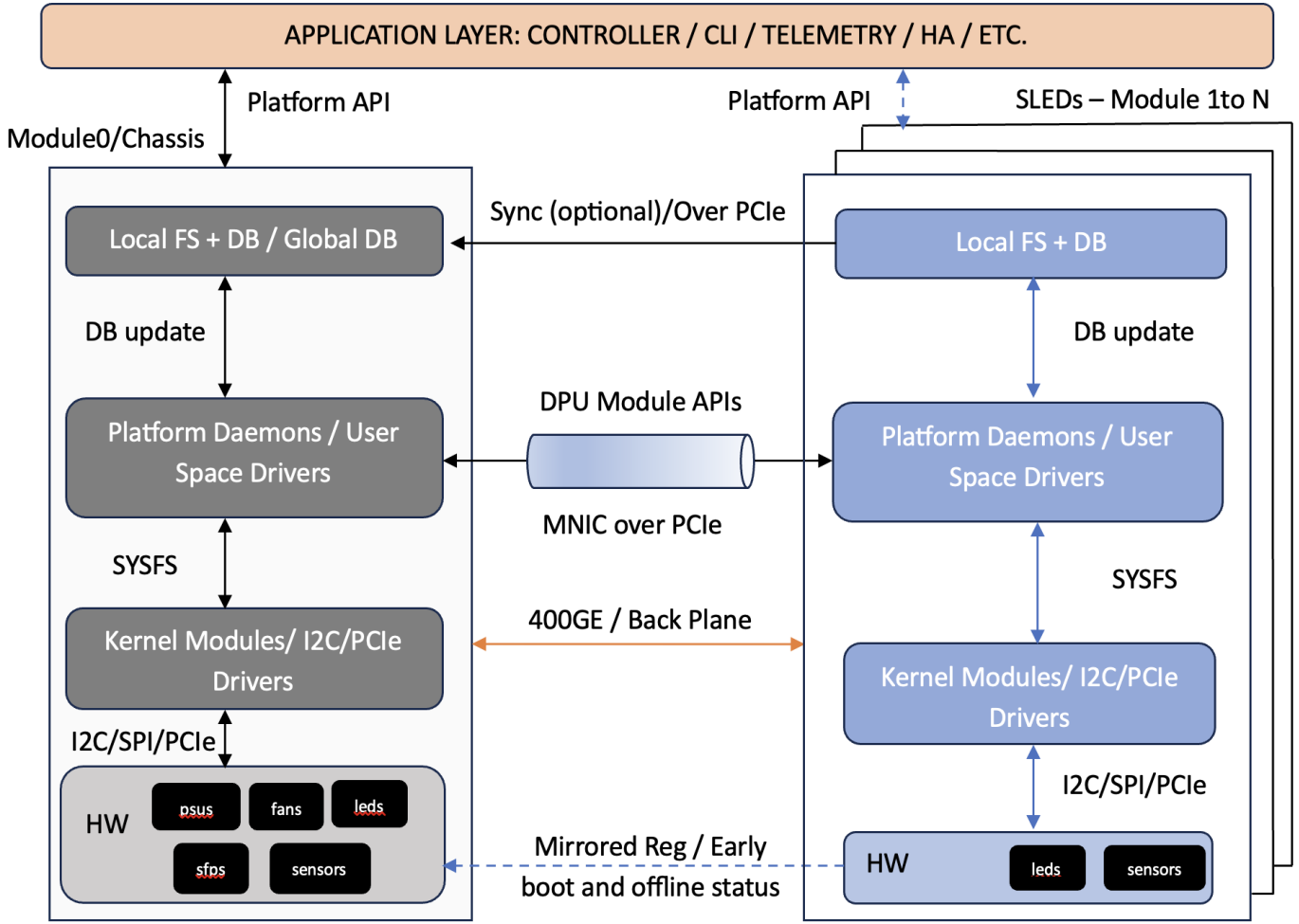


DPU Platform & LifeCycle Management

- Platform/Device Manager
- (lights off/on, BMC, fans, thermal, sensors)
- PCIe Virtualized-Ethernet/IPAM
- Firmware/BSP
- Upgrade/Image Mgmt
- Storage(eMMC)
- Sensors/last gasp
- OOB/BMC/Console
- Build & Packaging infra

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Modular chassis software stack



Each DPU sled is managed as a chassis module

01 Device lifecycle

02 Control plane

03 Data plane

PCIe Gen4 lanes partition to match platform needs

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Device lifecycle management

IDENTIFY

Identity, state, health and reboot cause

OPERATE

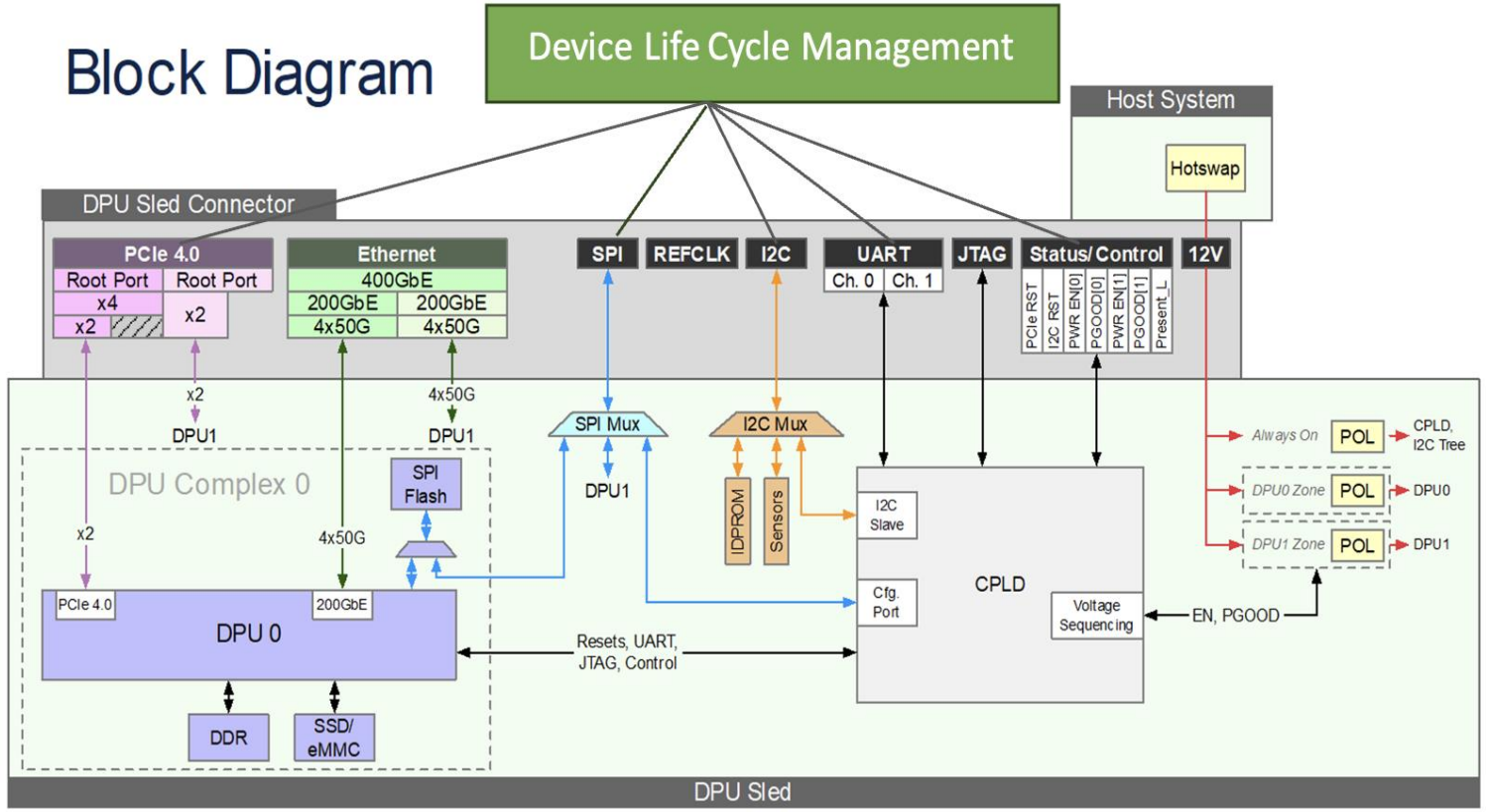
Reboot, power cycle and PCIe attach / detach

OBSERVE

Sensors, early boot state and console access

MAINTAIN

Firmware upgrade through a common interface



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Control plane connectivity

One virtual PCIe link per DPU

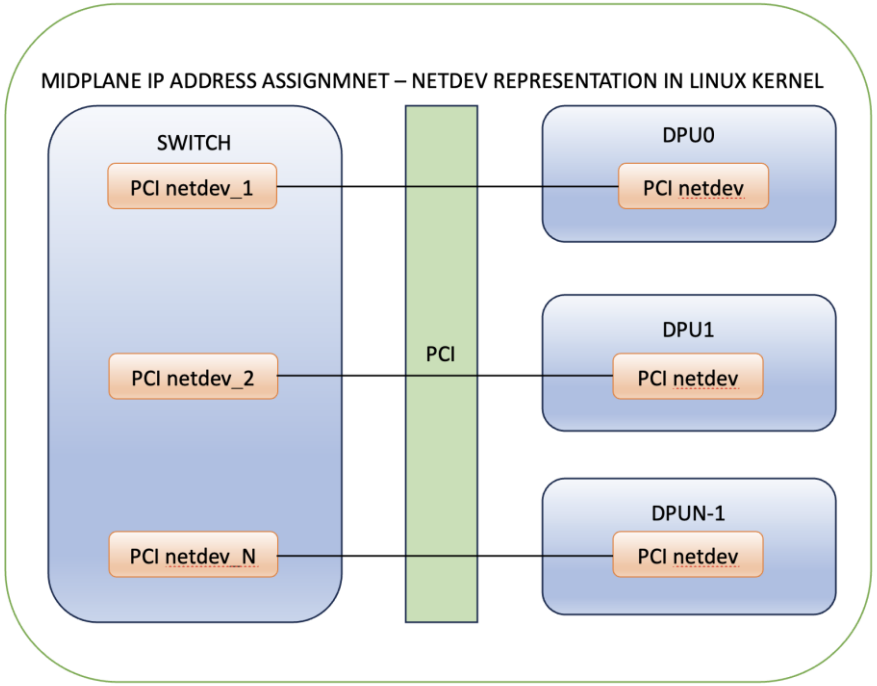
Each endpoint appears as a Linux netdev

CARRIES

State, health, configuration and image upgrade

STREAMS

Telemetry through gNMI / gRPC



Independent power control remains available per DPU

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Data plane connectivity

400GE

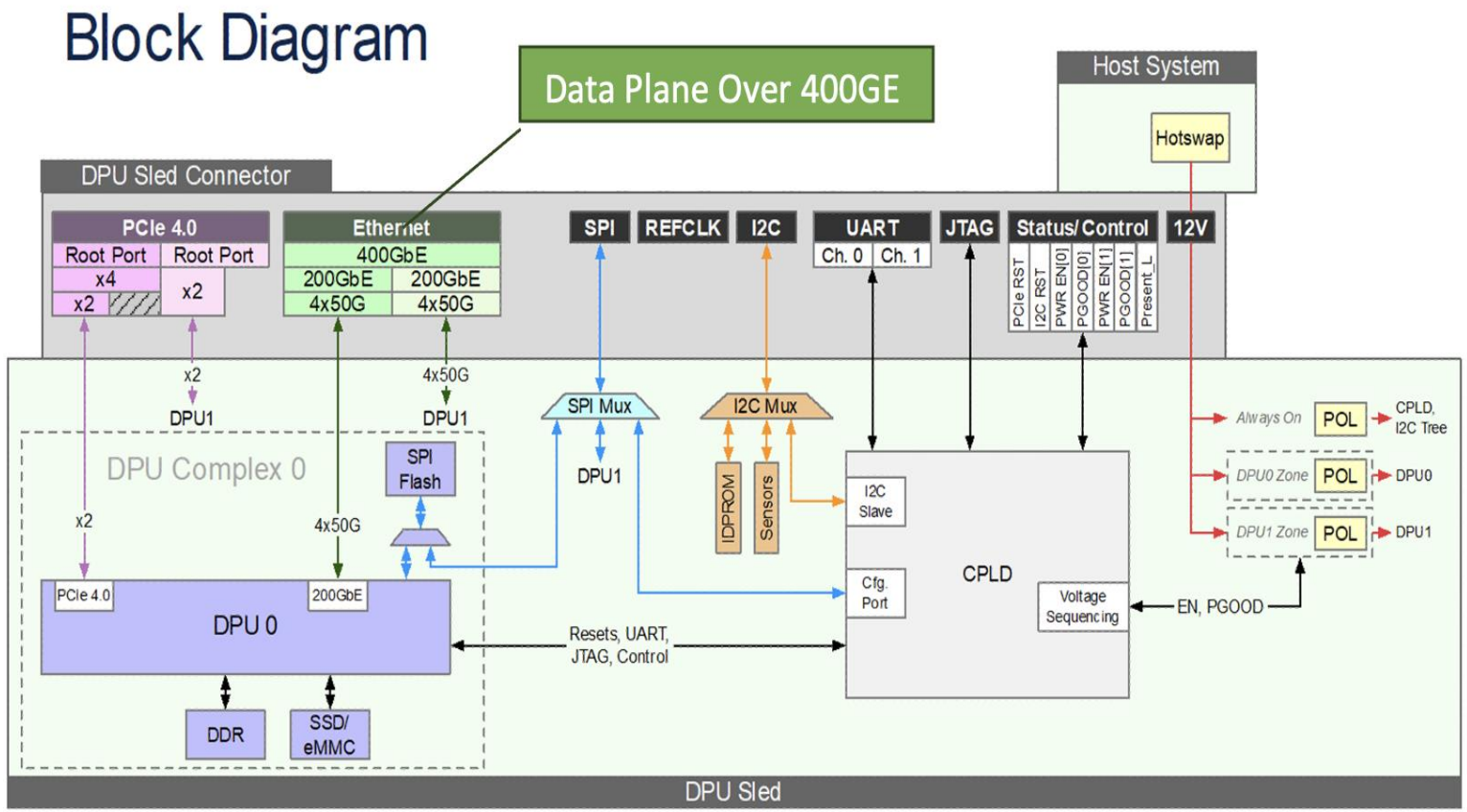
Data path on the sled connector

PROGRAMMABLE SERVICES

Pipeline offload, crypto, NAT and VPC

SECURE PLATFORM

Secure boot with QSPI and eMMC



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Next Gen Evolution of the Sled

Carrying forward the core principles, few Considerations for 100G/lane:

- Must support higher power levels to accommodate increased processing and data throughput demands
- Designed for higher data rates with improved signal integrity (SI), targeting 100G per lane or greater
  - Sled interface capable of supporting higher signaling rates
- Enhanced thermal capacity to manage increased power and data rates
- Improve Flexibility and Over-subscription ratios
- Rack-scale

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Call to Action

- Drive vendor agnostic nextgen hardware design in the eco-system for DPU/OEM/ODMs to accelerate development of DPU Sled
- Enable services builders to come up with ***new*** usecases/services
- Drive the standardization of software APIs and interfaces as necessary for Device Mgmt. and ***new*** Services
- Further details will be provided in the HW Spec to the workgroup
- JOIN *OCP-networking, help comment/develop the specification*

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Thank You!



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